

INTRODUCTION TO ECONOMICS

MGT404E

Single Unit Price Discrimination

Yale

Recap and Today's Plan

MGT404E Monopoly Pricing

Firms with market power are able to:

- Charge above marginal cost
- Earn positive profits

But, to sell an extra unit, they now face a tradeoff between higher quantity, but lower price on all units.

Thus, they choose quantity using $MR=MC$, or price using the Inverse Elasticity Pricing Rule:

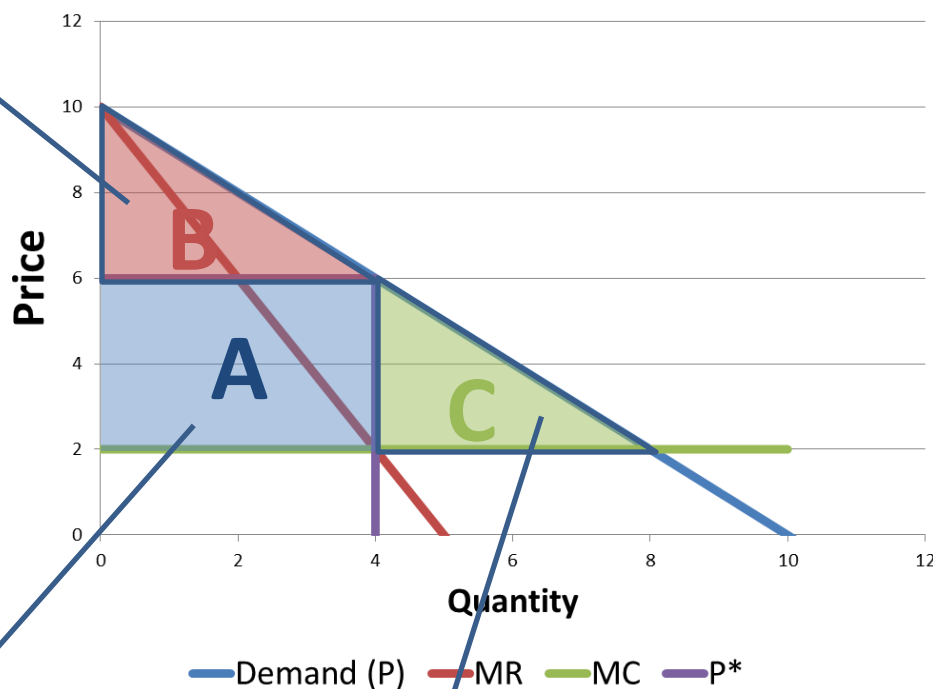
$$\frac{P^* - MC}{P^*} = \frac{-1}{\varepsilon}$$

However, this outcome is inefficient:

- There are consumers willing to pay above marginal cost, but below the firm's chosen price

Recall: Monopolist Pricing

Consumer
Surplus



Firm Profits

Deadweight
Loss (DWL)

Monopolist chooses a uniform price p to maximize profits.

This creates:

- Consumer surplus (B)
- Deadweight (C)

However:

- This leaves a lot of “money on the table”.

Price discrimination attempts to turn B and C into A. Is this good/bad?

Recap and Today's Plan

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Today: Begin pricing unit

The firm would love to capture some of the consumer surplus and deadweight loss as additional profit.

Strategy: Price Discrimination

- Involves charging different prices to different consumers
- Can increase profits and reduce deadweight loss

Firms price discriminate in many different ways:

- By different individuals (perfect, demographic, and self-selecting)
- By charging differently for different quantities (quantity discounting, membership)
- And, combinations: menu pricing and two-part tariffs

today

next
class

Motivation

Uniform Pricing

So far, we saw that a firm with market power can choose a single price to maximize profits. In reality, we see firms offering **the same goods at different prices to different people**.

Examples of Price Discrimination in Practice

- Airline tickets: advance purchase, Saturday night stay-over, etc
- Cell phones, Electricity: off-peak/on-peak pricing
- Others?



When is this a good strategy? What are the different approaches to price discrimination?

The Happiest place on Earth



The Happiest place on Earth



- Historically, Disney charged the same price any day for admission to the park.
- **Does this make sense?**
- Would your reservation price to go to Disney theme parks be the same year round? What would affect it?
 - Weather? Time of year? Day of week?
 - (... crowds?)
- What are some consequences of uniform pricing?

When to go to Disney:



WDW | December 2025

CROWD CALENDAR						
Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

- Quiet as a Mouse - Few and far between, these days are gold!
- Less Busy than Average - Lucky ducks will enjoy lighter crowds on these days.
- Average Crowds - A pretty regular day in Walt Disney World.
- More Busy than Average - Heavier crowds on these days make planning essential.
- Wall-to-Wall People - You must be a people person to visit on the most crowded days!

Data from dadsguidetowdw.com

Epic photobomb:



Two ways to price discriminate

Consumers have demand for...

A Single Unit

- Different individuals have different reservation prices
- Cannot necessarily charge everyone their exact reservation price (resale? Arbitrage?)
- Could use observable **demographics** to price.
- Could try to get “high type” consumers to opt-in to paying more. (Self-selecting pricing)

Multiple Units

- An individual values additional units of a product less than they value the first unit
- Ideally charge differently for additional units
- Can try “**membership pricing**” – unlimited units
- Can also try self-selecting:
 - Can offer a **menu** of quantities
 - Can charge a **two-part tariff**.

Example: Hungry Late-Night Students



Vending machine, same products, lower price on higher shelf.

- Lower price items sell out during the day
- Late at night, few alternatives available, willingness to pay is higher... and only the more expensive shelf remains stocked.

(this photo was not taken at Yale!)

Via Freakonomics blog: <http://www.freakonomics.com/2010/01/15/freak-shots-honest-mistake-or-snack-gouging/>

This one is for those who speak Spanish:



A nice little variation on the
“locals discount”

When Can You Price Discriminate?

The following are necessary conditions for successful price discrimination:

Condition	Why?
1. Market Power	• A firm without market power cannot set a price different from the market price and still achieve positive sales.
2. Consumers Valuations Differ	• If all consumers agree on the value of the good, there is no surplus to be captured above that value.
3. Resale is Not Possible	• If consumers able to purchase at a lower price could resell to consumers offered a higher price, the first could profitably resell the goods (arbitrage)

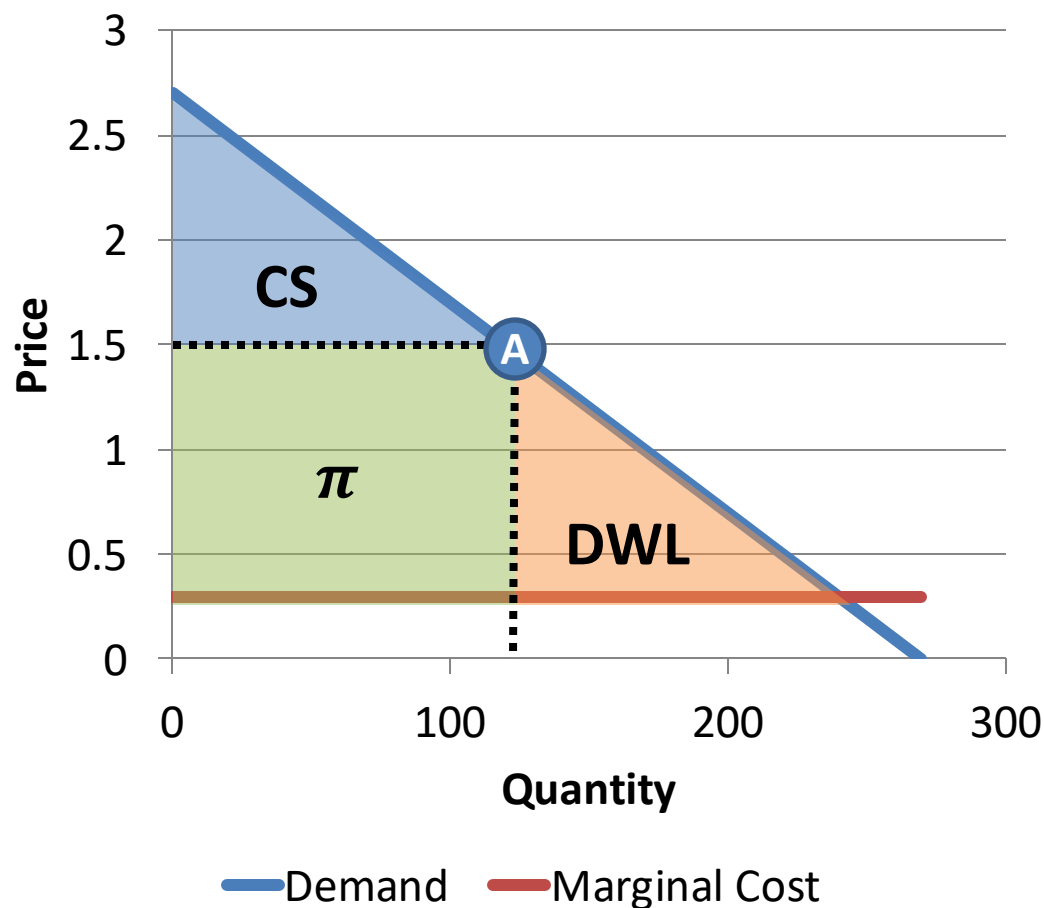


Price discrimination is not always possible
When are these conditions more or less likely to hold?

Example: Safeway Bakery

Example: The Safeway Bakery

The Safeway Bakery is choosing a uniform price for products it sells. Given an estimated demand curve, they can choose the price for a loaf of bread.



Optimal uniform price:

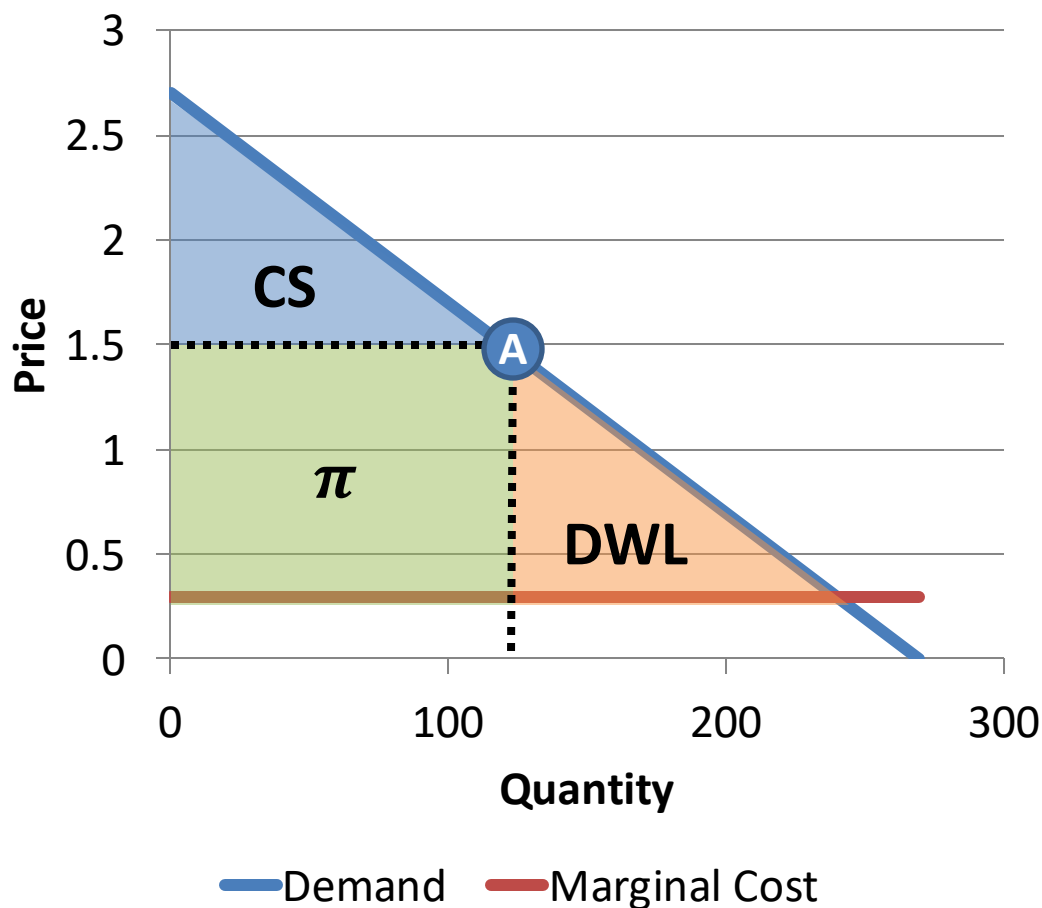
Given this demand curve, a price of \$1.50 maximizes profits.

At this price, the firm captures profits, some consumers capture surplus, and some deadweight loss is created.

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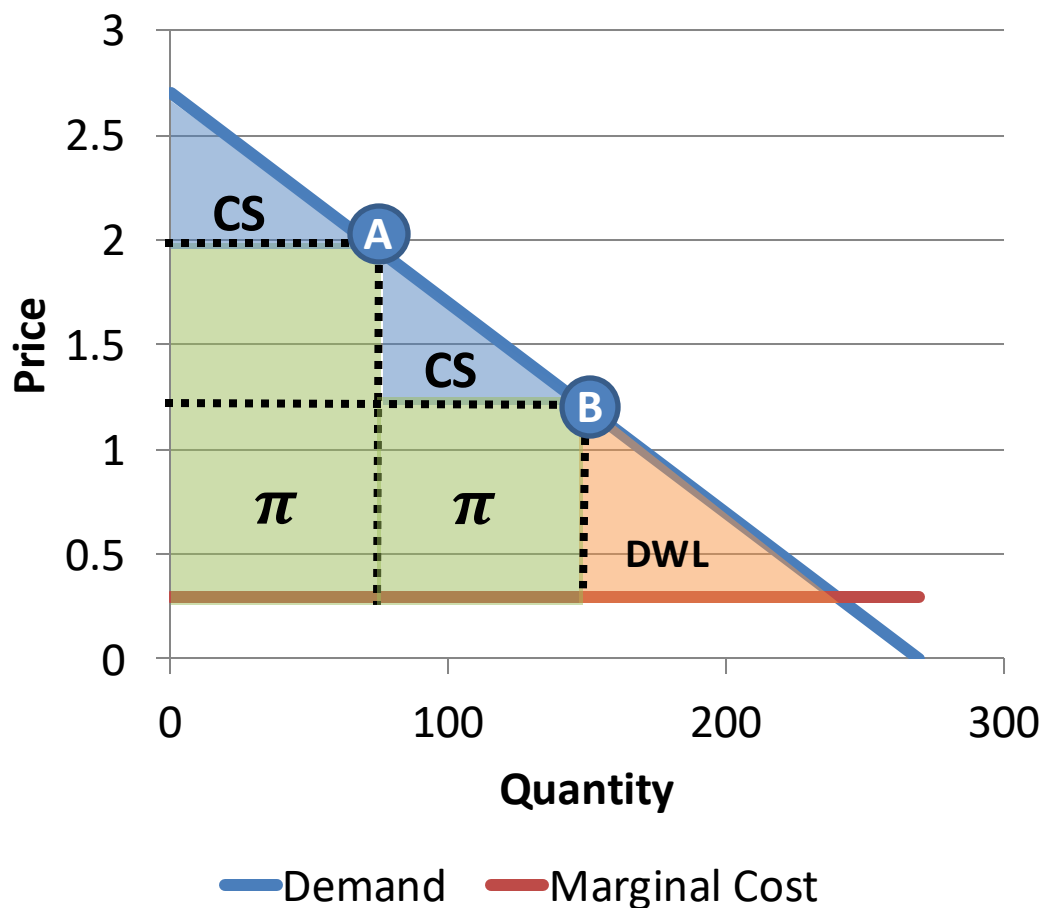
Some price discrimination...

First, Safeway could offer lower prices (\$1.25) to those consumers who it knows value the bread below \$1.50 and higher prices (\$2) to those who value it above \$1.50. This will allow them to capture some of the DWL as profit.

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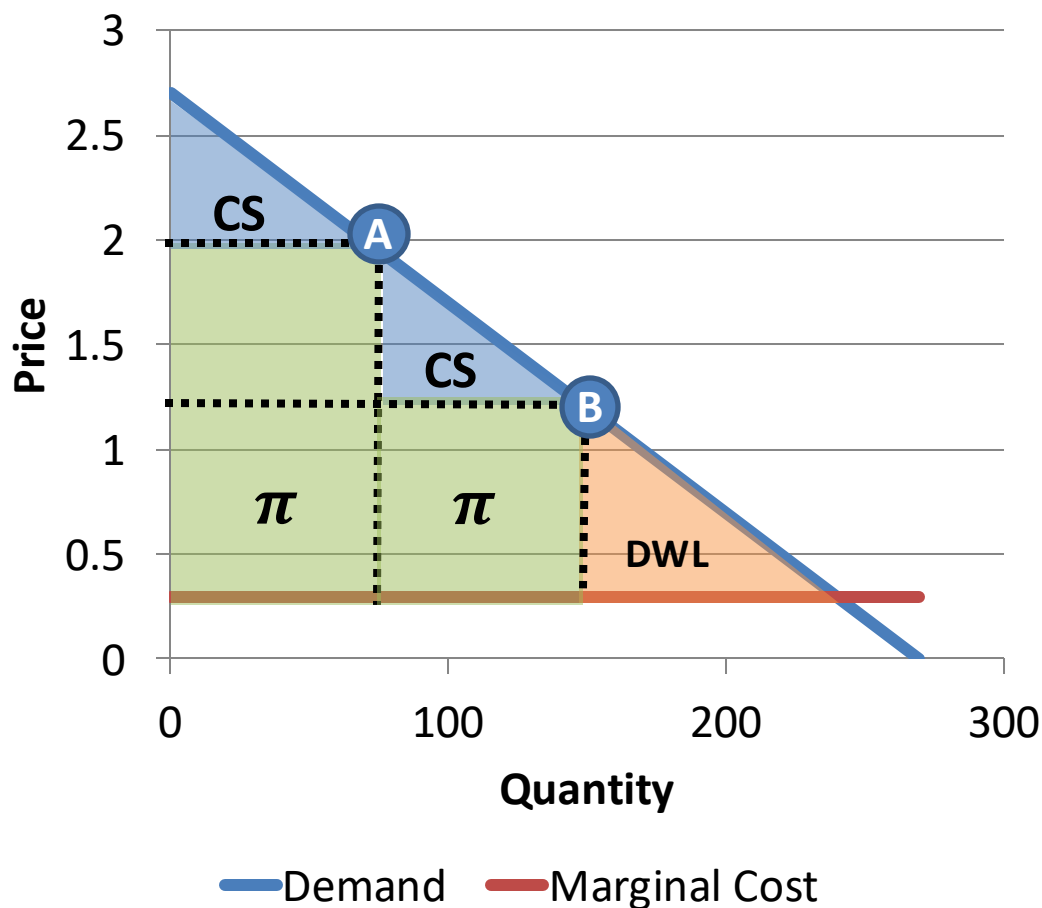
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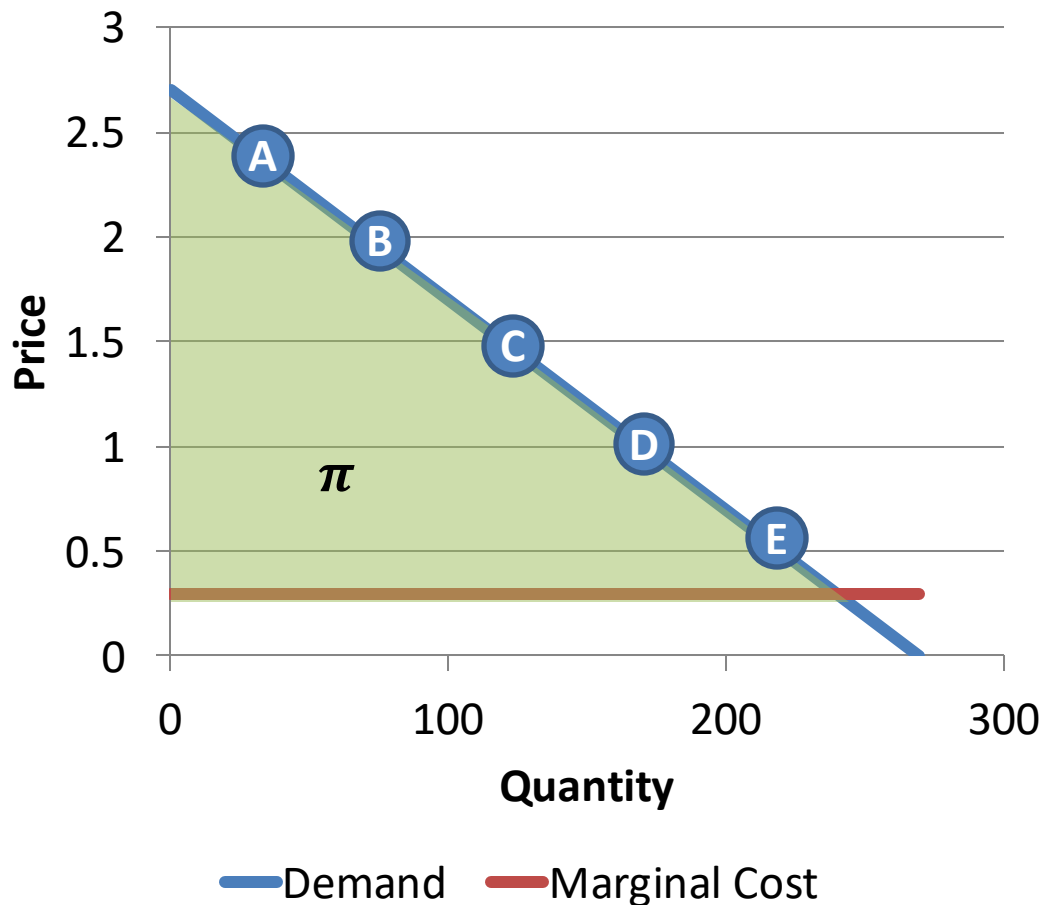
Perfect price discrimination

Best case scenario: If the firm knows each consumer's valuation, it can charge each of them exactly their valuation as price and capture all surplus created in the market.

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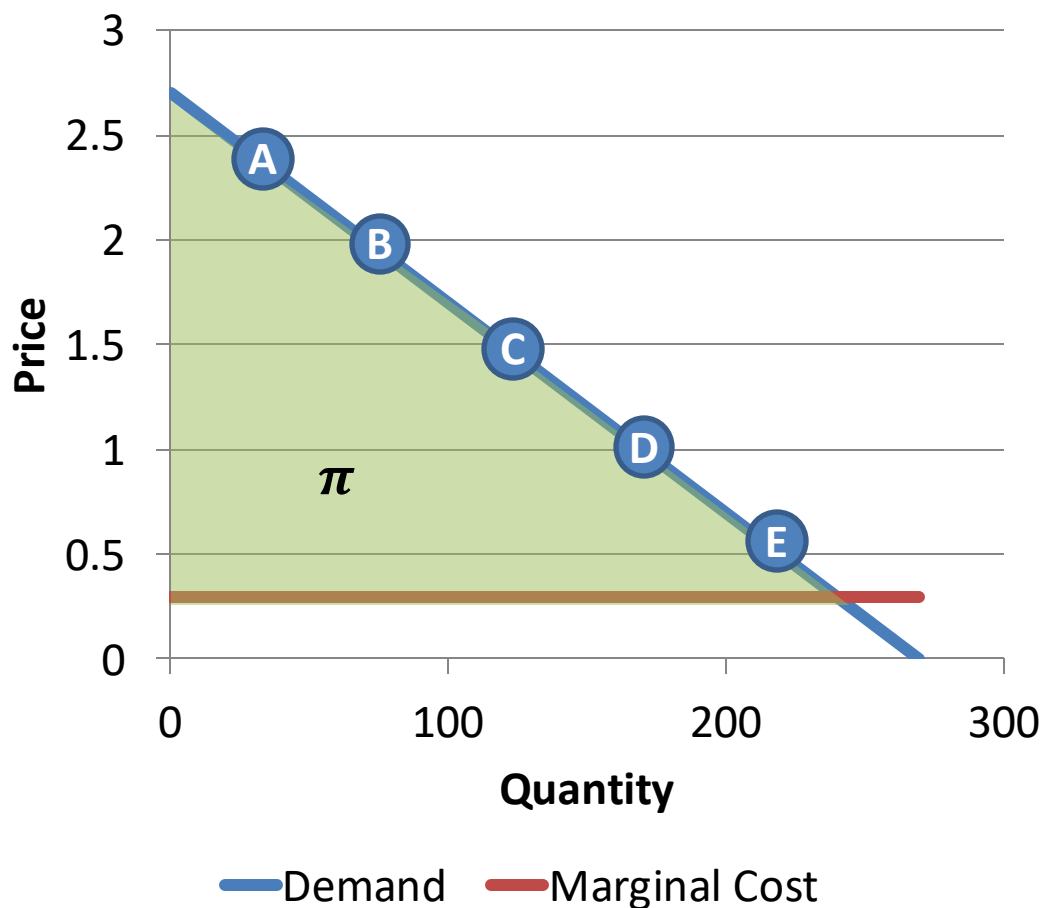
Perfect price discrimination

Best case scenario: If the firm knows each consumer's valuation, it can charge each of them exactly their valuation as price and capture all surplus created in the market.

How do we feel about this?

Example: The Safeway Bakery

The Safeway Bakery is choosing a uniform price for products it sells. Given an estimated demand curve, they can choose the price for a loaf of bread.



“Ethical” implications

- Safeway selling to consumers in the DWL region seems good—it is more efficient, and consumers get the product they want at a lower price.
- On the other hand, Safeway using price discrimination to cut into the consumer surplus seems to harm consumers.
- Is there really a difference? We generally think that capturing surplus – by anyone – is better than letting surplus go to waste.

Types of Single-Unit Price Discrimination

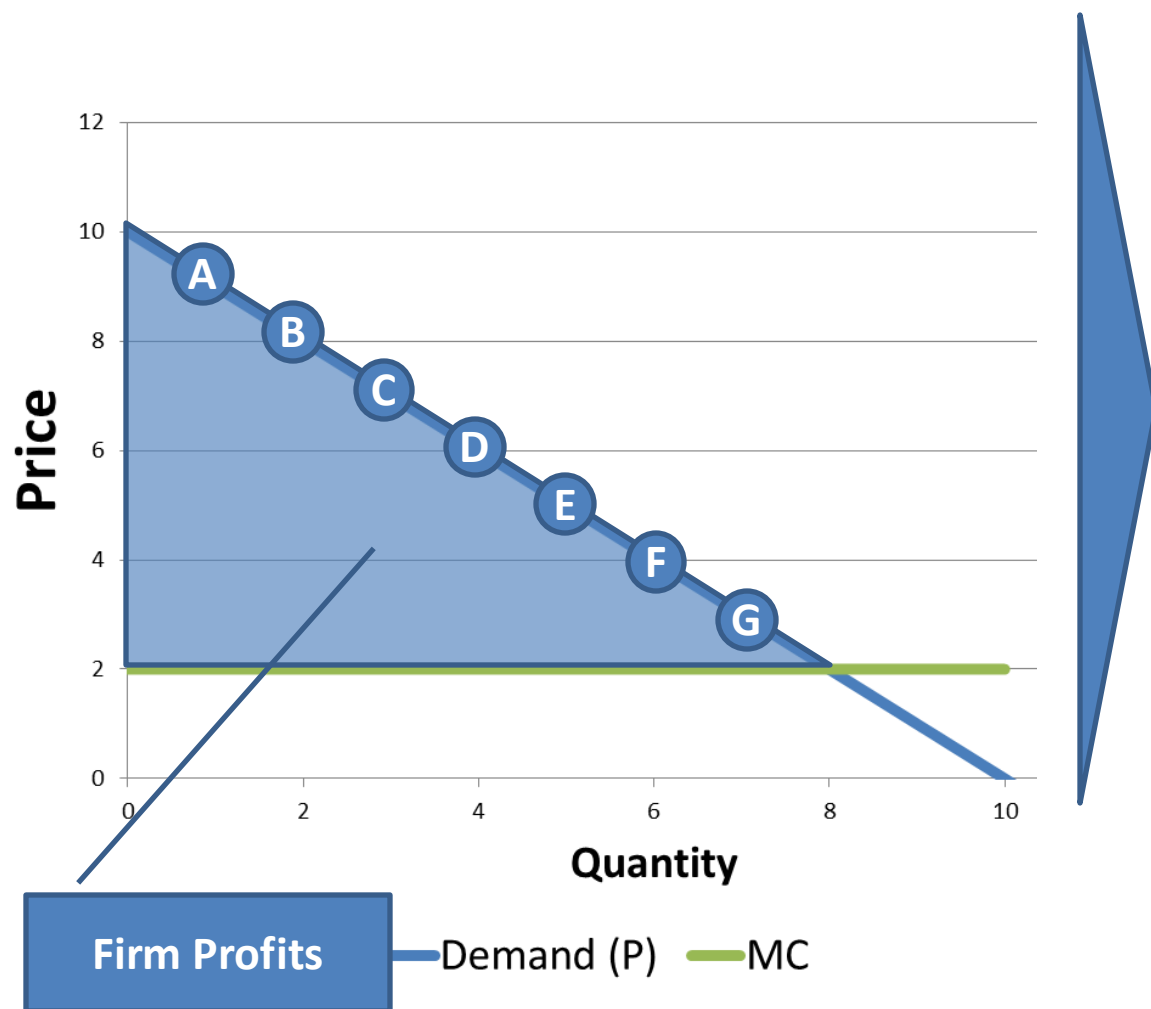
Maintained Assumptions

In all cases, the firm has **market power**, consumers have **different valuations** of the good, and **resale** is not possible.

Type	What the firm knows	Strategies to capture surplus
Perfect	• Each consumer's exact valuation	• Set individual prices
Demographic	• A consumer's "type", which is correlated with their valuation	• Offer unique prices to identifiable groups
Self-Selecting	• There are different types of consumers who value the good differently	• Look at incentives of different types; design packages for each.

Strategy for capturing additional surplus through price discrimination depends on what a firm knows about consumers

Perfect Price Discrimination



Perfect Price Discrimination:

- The firm knows each consumer's reservation price
- Charge each consumer exactly their reservation price

Result: Firm captures all possible surplus

- No deadweight loss, efficient outcome
- If re-sale were possible, this would not work. What else makes it impractical?

Examples of perfect price discrimination?

Yale College Financial Aid

2021–2022 Estimated Cost of Attendance for Yale Undergraduates

Tuition and Fees	\$59,950
Room (Housing)	\$10,100
Board (Meal Plan)	\$7,700
Estimated books and personal expenses	\$3,700
Student Activities Fee	\$125
TOTAL	\$81,575

Class of 2023 First-Year Students

Annual Income Range	Median Net Cost*	Median Scholarship	Percentage Who Qualified for Aid
Less than \$65,000	\$2,850	\$76,925	100%
\$65,000-\$100,000	\$5,701	\$70,217	99%
\$100,000-\$150,000	\$15,528	\$60,295	99%
\$150,000-\$200,000	\$29,721	\$46,326	95%
\$200,000-\$250,000	\$42,964	\$31,196	79%
Greater than \$250,000**	\$43,704	\$28,881	25%

64%

Undergraduate students who receive financial assistance (2017-2018)

\$0

Parent contribution for families with less than \$65,000 in annual income and typical assets

\$51,500

Average Yale need-based scholarship (2018-2019)

\$12,500

Median net cost for students receiving financial aid (2017-2018)

This is Price Discrimination: Remember, ability to pay is necessary for “willingness” to pay—Yale charges those with higher reservation prices more.

Example of Perfect Price Discrimination

College Financial Aid

Setting

- Colleges have market power: enrollment responds to changes in tuition.
- Tuition itself is priced uniformly.
- Schools typically offer both merit-based aid, as well as need-based aid.
- Students submit financial data to be considered for financial aid.

Financial Aid

- Financial data indicate ability to pay for the education.
- Colleges make offers of financial aid based on need, lowering the price of the education on an individual basis
- The good cannot be transferred: the offer is tied to an individual student.

Financial aid increase efficiency by enabling students willing to pay above the marginal cost of their education, but not willing (able) to pay full tuition, to enroll.

But, is it only that? Or are they also using it to eat into “consumer surplus” at the top?

Challenges with Perfect Price Discrimination

Challenge	Strategies
Preventing Resale	• Works best with services (which cannot typically be transferred) • Goods that are difficult to transfer (cars, large equipment) or whose value is connected to the individual.
Knowing Individual Reservation Prices	• Almost impossible to know – consumers may not know themselves • Look for individual characteristics correlated with reservation prices



Very difficult to achieve in practice.

Electronic Price Tags, our future?



- Useful for purposes such as reducing “menu costs”
- However: huge potential for personalized pricing
- Same applies for checkout coupons based on loyalty cards

Types of Single-Unit Price Discrimination

Maintained Assumptions

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Strategy for capturing additional surplus through price discrimination depends on what a firm knows about consumers

Example: Train Discounts

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DESTINATIONS | EXPERIENCE | DEALS | SCHEDULES | GUEST REWARDS |  TRAIN STATUS |  MODIFY TRIP

One-Way | Round-Trip | Rail Passes | Auto Train | Other Options ▾

Use Points ☐

From  To Depart Date — Return Date

Advanced Search ▾

Travelers 
[Clear](#)

Adult
16+

- 1 +

Youth
13 - 15 | [Traveling Alone?](#)

- 0 +

Child
2 - 12

- 0 +

Infant
Under 2

- 0 +

Discounts ⓘ

Traveler 1: Adult

Adult 

Senior

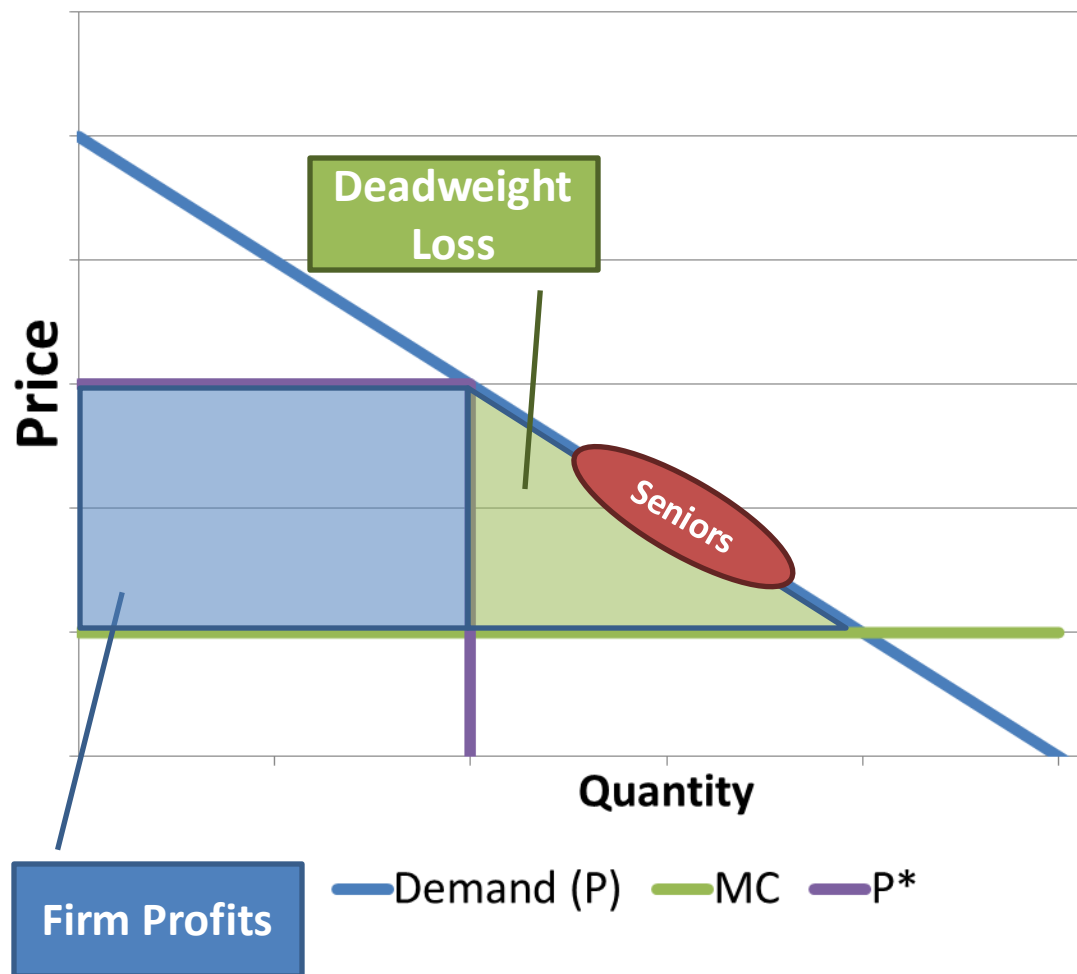
Passenger with Disability

Companion



- Amtrak offers a lower price to seniors
- To qualify, must offer **proof of status**
- Tickets are non-transferrable (no resale)

Targeting Segments with Different Valuations: Amtrak



At the optimal uniform price, seniors are not buying train tickets.

- By offering senior discounts, the firm captures additional profits that would otherwise be deadweight loss.
- Because seniors must present proof of status and tickets are tied to individuals, other purchasers still pay the original price
- This increases firm profits and overall efficiency

Mathematical Example: Amtrak

The Situation

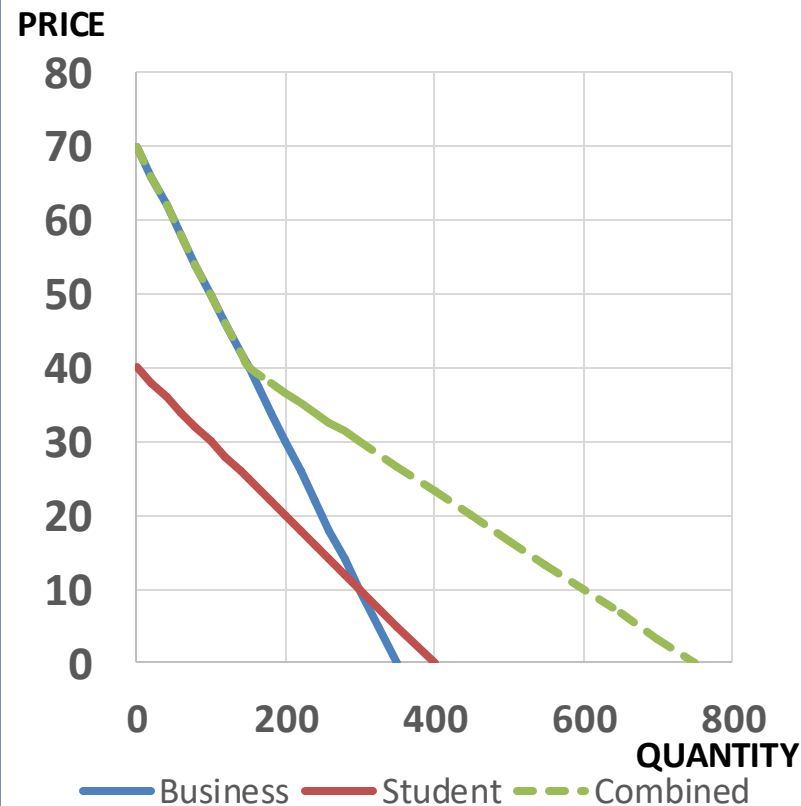
You are in charge of pricing Amtrak's route from New Haven to New York.

- Let Q be the quantity
- There are two main types of customers for this route: Business, and Seniors.
- Demand among Business travelers is given by $P_B(q) = 70 - 0.2q$
- Demand among seniors is given by $P_S(q) = 40 - 0.1q$
- Marginal cost for both groups of travelers is constant at \$4 (ticketing, fuel, etc).

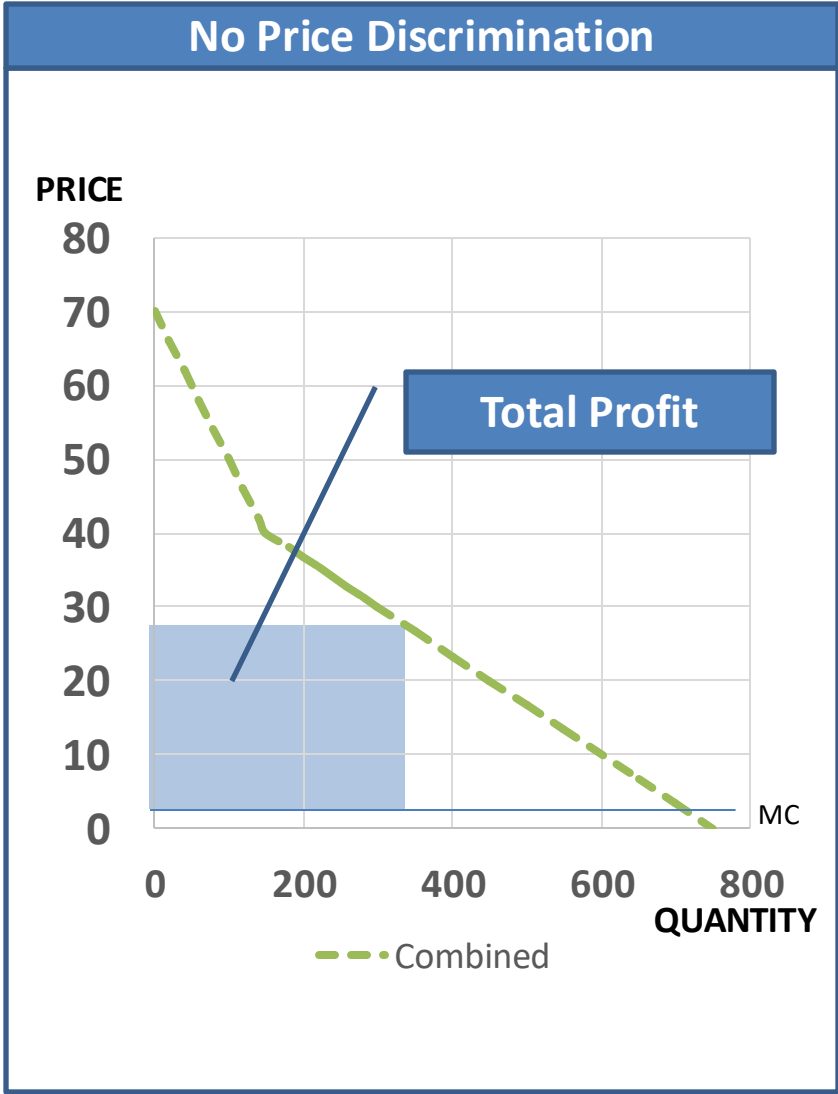
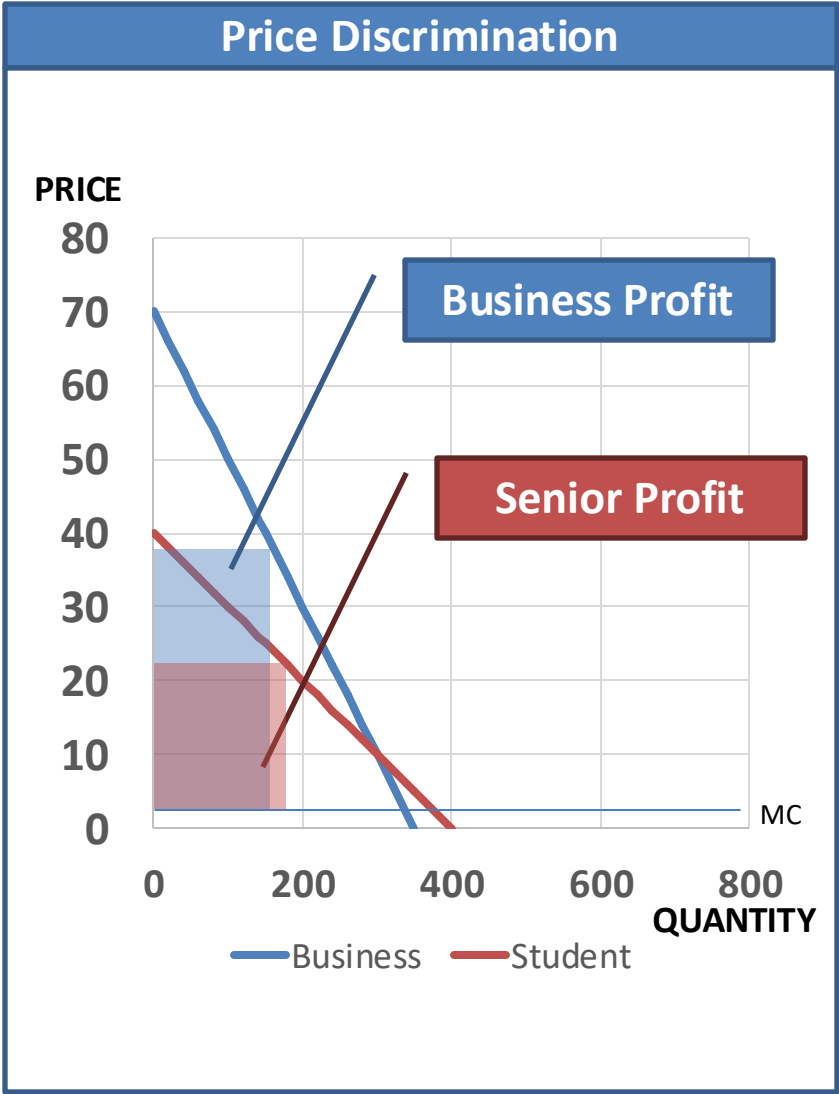
How much can Amtrak increase profits if it is able to price to each group separately?

Demand

Recall how to aggregate demand from different types of customers.



Amtrak: Graphic



If Amtrak is able to price discriminate

...then it sets a different price for each market segment (by setting $MR=MC$):

Business: $P_B(q) = 70 - 0.2q$

Write out Revenue(q) and $MR(q)$:

$$\begin{aligned}R_B(q_B) &= q_B \cdot P_B(q_B) \\&= 70q_B - 0.2q_B^2 \\MR &= 70 - 0.4q_B\end{aligned}$$

Set $MR=MC$:

$$\begin{aligned}70 - 0.4q_B^* &= 4 \\q_B^* &= 165\end{aligned}$$

Hence,

$$\begin{aligned}p_B^* &= P_B(q_B^*) = \$37 \\ \pi_B(q_B^*) &= (\$37 - \$4) \cdot 165 \\ &= \$5,445\end{aligned}$$

Senior: $P_S(q) = 40 - 0.1q$

Write out Revenue(q) and $MR(q)$:

$$\begin{aligned}R_S(q_S) &= q_S \cdot P_S(q_S) \\&= 40q_S - 0.1q_S^2 \\MR &= 40 - 0.2q_S\end{aligned}$$

Set $MR=MC$:

$$\begin{aligned}40 - 0.2q_S^* &= 4 \\q_S^* &= 180\end{aligned}$$

Hence,

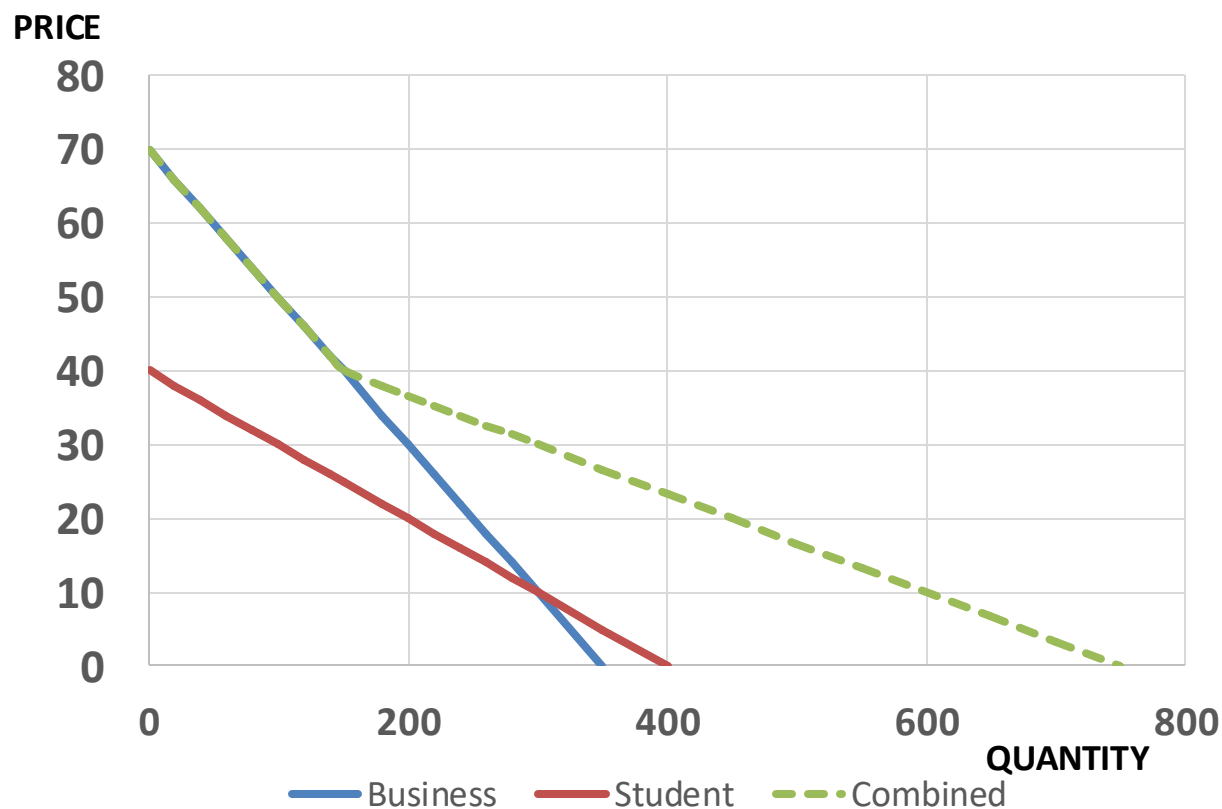
$$\begin{aligned}p_S^* &= P_S(q_S^*) = \$22 \\ \pi_S(q_S^*) &= (\$22 - \$4) \cdot 180 \\ &= \$3,240\end{aligned}$$

So, when Amtrak price discriminates, it makes \$8,685 (= \$5,445 + \$3,240).

If Amtrak cannot price discriminate

Demand

Recall how to aggregate demand from different types of customers.



If Amtrak cannot price discriminate

First, aggregate demands by expressing demand as $q(P)$

$$\begin{aligned}P_B(q) &= 70 - 0.2q & Q_B(p) &= 350 - 5p \\P_S(q) &= 40 - 0.1q & Q_S(p) &= 400 - 10p\end{aligned}$$

Now, Amtrak will need to check both sides of the kink, that is, check whether it wants to serve both types, or just businesspeople.

Serving only business: $p \geq 40$

Spoiler: we already solved that the best price for business is \$37, so serving only business doesn't make sense.

Serving both types: $p \leq 40$

Aggregate demand:

$$\begin{aligned}Q(p) &= Q_B(p) + Q_S(p) \\&= 750 - 15p \\P(q) &= 50 - 1/15 q \text{ (if } p < 40\text{)}\end{aligned}$$

Write out Revenue, Marginal Revenue:

$$\begin{aligned}R(q) &= 50q - 1/15 q^2 \\MR(q) &= 50 - 2/15 q\end{aligned}$$

Set $MR=MC$:

$$\begin{aligned}50 - 2/15 q &= 4 \\q^* &= 345\end{aligned}$$

Plug in optimum to get price and profit:

$$\begin{aligned}p^* &= P(345) = \$27 \\ \pi &= (\$27 - \$4) \cdot 345 \\ &= \$7,935\end{aligned}$$

Amtrak: Profits from Price Discrimination

If Amtrak cannot price discriminate...

Maximize profits subject to aggregate demand:

- Set a uniform price of \$27
- Total quantity of tickets: 345 (130 senior tickets, 215 business tickets)
- Profits: $\pi = (\$27 - \$4) \cdot 345 = \$7,935$

If Amtrak is able to price discriminate...

Maximize profits separately for each group:

- Set a senior price of \$22 and a business price of \$37
- Total quantity of tickets: 345 (now 180 senior tickets, 165 business tickets)
- Profits: $\pi = (\$22 - \$4) \cdot 180 + (\$37 - \$4) \cdot 165 = \$8,685$

- If Amtrak had to price to the entire market, the more price-sensitive seniors “drag down” the optimal price. When they can discriminate between seniors and business (e.g., senior IDs), they can capture additional surplus from business travelers.
- Notice the gain is only \$750—what if there was a cost of implementing this scheme, such as some business travelers cheating?

Types of Single-Unit Price Discrimination

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Demographic	A consumer's "type", which is correlated with their valuation	Offer unique prices to identifiable groups
Self-Selecting	There are different types of consumers who value the good differently	Look at incentives of different types; design packages for each.

Strategy for capturing additional surplus through price discrimination depends on what a firm knows about consumers

What if you cannot verify Demographics?

You Are an Airline	Setting Prices	Result
- You are setting the price of the New York – Los Angeles route - There are two types of travelers: business, and leisure. Business travelers are willing to pay more than leisure.	- You calculate the optimal price for business travelers (“high types”) is \$400. For the leisure travelers (“low types”), it is \$150. - Tickets are put up for sale: \$400 for business travel, \$150 for leisure.	- Business travelers would just say they are traveling for leisure! - The airline has no way to verify or enforce the segmentation.

- Instead, you impose an “early purchase requirement” for the Leisure fare (must be purchased at least 3 weeks before travel).
- Business travelers then “**self-select**” into the business ticket, since they are less likely to have planned the travel that far in advance.
- Additional approaches: Saturday night stayover, “flexible fares”, Business Cabin, etc.

Introduction to Self-Selecting Price Discrimination

Key Fact

Consumers have different reservation prices, and you would like to charge them according to their willingness to pay. “High types” -> high reservation prices.

But

You can't split them into enforceable demographic groups, or you worry about making them angry with “discriminatory” pricing

What You Do Instead

Design your product so that consumers self-select into different prices. Buyers “opt-in” to higher prices. The trick? Price according to product characteristics correlated with reservation prices.

Examples of Self-Selecting Price Discrimination

- “Top of the Line iPad” – Some people get value from getting the “best” available, and so will choose the higher priced product
- Advance purchase requirements, or having flexible tickets only with one fare class – needing these things is related to having higher willingness to pay
- Making people jump through hoops for deals – higher valuation people have more money, less time. (e.g. coupons)
- Others?

Introduction to Self-Selecting Price Discrimination

The Downside

- You don't get to extract all consumer surplus, except from the "low" types.
- The higher "type" consumers (higher reservation prices) will earn what economists call "information rents" from you. You will leave them some consumer surplus and in exchange, they will pay more for the "high type" product.

We will now set prices for a "High type" good and a "Low type" good

The "Willing to Buy" Constraint

- Each type must get non-negative consumer surplus, so they'll purchase at all.
- Most important for "low types".

(Economics jargon: Individual Rationality Constraint)

The "Buy the Right One" Constraint

- You are concerned the "high types" will get more surplus from the "low type" product, since it is cheap.
- So you need to "leave some money on the table" (consumer surplus) so they're happy about choosing the higher priced good.

(Economics jargon: Incentive Compatibility Constraint)

Example: Golf Course Pricing

Setting

- You run a golf course, and you just know that different types of golfers with different valuations. But, there is no verifiable trait you can use to explicitly discriminate based on.
 - You decide to try to create self-selecting pricing, where you have different prices for golfing in different situations.
- 

What are some options?

- Time of day / day of week
- Perks
- Groups versus alone

Golf Course Example

Setting

- You do a little market research and find out there are: “flexible” and “weekender” types who golf on your course. **People want to golf at most once per week.** Flexible golfers like golfing on the weekend, but only a little bit more than weekdays. Whereas weekenders really value being able to golf on the weekends.
- At the same time, the weekenders seem to value golf more overall, whereas the flexible ones seem to value it less.

Consumer Reservation Prices:

	“Flexible” (100ppl)	“Prefer Weekend” (100ppl)
Weekday	\$40	\$50
Weekend	\$50	\$80

Assume my marginal cost is a constant \$20 for all days, and people are only interested in golfing once a week. What’s my strategy?

- Can’t charge different prices by type—why would people tell me their “type”?
- Should I just set $P=\$80$ at all times? Should I set $P=\$50$ at all times?
- Or can I do better?

\$6000 profit
(\$60x100 ppl)

\$6000 profit
(\$30 x 200ppl)

Example Continued

Consumer Reservation Prices:

	"Flexible" (100ppl)	"Prefer Weekend" (100ppl)
Weekday	\$40	\$50
Weekend	\$50	\$80



- You cannot price to different individuals, but you **can** charge a different price on weekend vs weekday.
 - Suppose you set $P_{weekday} = \$40$ and $P_{weekend} = \$70$
 - Flexible will choose to golf during the week
 - "Prefer Weekend" will choose to golf on the weekend
 - Profits are $\$20 \times 100 + \$50 \times 100 = \$7,000$, a 17% increase.

Question: Why not set $P_{weekend} = \$80$?

- The weekend types would then rather pay \$40 and golf during the week. They would get more surplus from weekday golfing. They earn \$10 of surplus as "information rents".

Example: How to Come up with Prices

Consumer Reservation Prices:

	“Flexible” (100ppl)	“Prefer Weekend” (100ppl)
Weekday	\$40	\$50
Weekend	\$50	\$80



To set the right prices, we need to use our two constraints

1. “Willing to Buy” Constraint:

- Will the low type (Flexible) type buy the “weekday” product?
- Need $\$40 - P_{weekday} \geq 0$ for “flexible” (Low type is the one to check)

2. “Buy the Right One” Constraint:

- Will the high type (Prefer Weekend) buy the weekend over the weekday product?
- Need $\$80 - P_{weekend} \geq \$50 - P_{weekday}$ (High type is the one to check)
- Thus if I charge the most I can for weekday golfing, \$40, I need to leave at least \$10 of “value” for the weekenders, so the most I can charge is \$70
- → Hence the prices of \$40 and \$70 are the best.

Self-Selecting Pricing in General

In general, to find the best self-selecting scheme:

- Charge the low type exactly their reservation price. This ensures you meet the “willing to buy” constraint.
- Charge the high type the maximum that you can, making them indifferent between “high” good than “low” good.



Then compare with *best* single price, by calculating profit for each scheme. Also, costs can be different between the goods.

Back to Golf Example: High/Low Type and Quality

Consumer Reservation Prices:

	"Flexible" (100ppl)	"Prefer Weekend" (100ppl)
Weekday	\$40	\$50
Weekend	\$50	\$80

"Low Type" Consumer

"High Type" Consumer



"Low Quality" Good

"High Quality" Good

1. Find best self-selecting scheme

- Charge the low-type reservation price for the low-quality product: Weekday Price is \$40
- Charge the highest price for the high-quality good such that the high-types are willing to buy it: Weekend Price is \$70
→ Profit=\$7,000

2. Compare to best single price scheme

- Either charging \$80 (and getting only weekenders) or charging \$50 and getting both types yield profit of \$6,000. So in this case we want to price discriminate.

Price discrimination not always possible

- Note – in this case, you screen, but in other cases, you may not! I.e. you may pool.

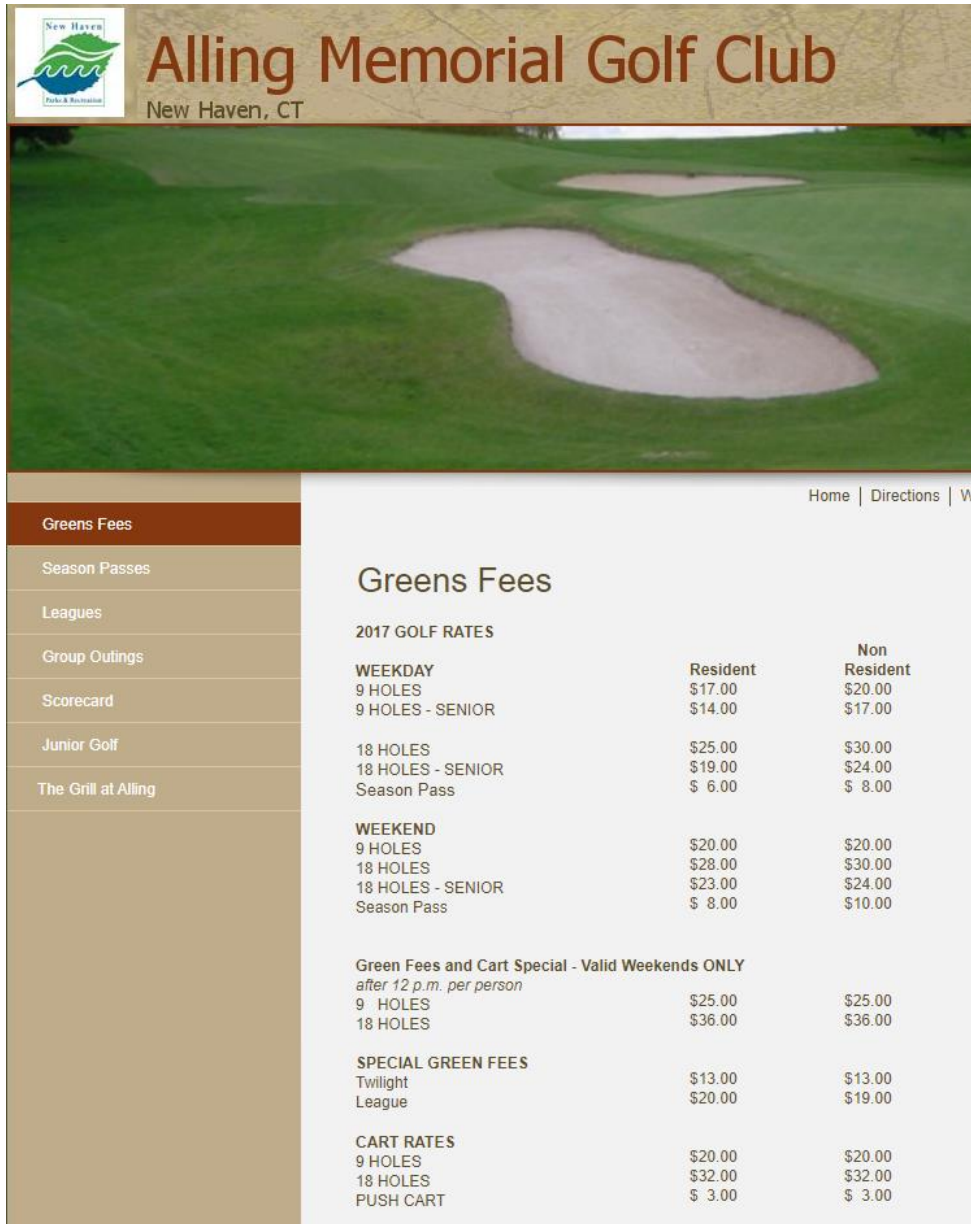
What if consumer reservation prices were:

	“Flexible” (100ppl)	“Prefer Weekend” (100ppl)
Weekday	\$40	\$50
Weekend	\$60	\$70

1. “Willing to Buy” Constraint:
 - Need $\$40 - P_{\text{weekday}} \geq 0$ for “flexible” (Low type is the one to check)
 2. “Buy the Right One” Constraint:
 - Need $\$70 - P_{\text{weekend}} \geq \$50 - P_{\text{weekday}}$ (High type is the one to check)
- We end up with $P_{\text{weekday}} = \$40$ and $P_{\text{weekend}} = \$60$.
 - But at these prices, everyone comes on weekends!

\$8000 profit
(\$40x200 ppl)

Sure enough:



The screenshot shows the website for Alling Memorial Golf Club in New Haven, CT. The header features the club's logo and name. Below the header is a large image of a golf course with a sand trap. A navigation menu on the left lists various options: Greens Fees, Season Passes, Leagues, Group Outings, Scorecard, Junior Golf, and The Grill at Alling. The main content area displays the 'Greens Fees' section, which includes a table of 2017 Golf Rates. The table is organized by category (Weekday, Weekend, Green Fees and Cart Special, Special Green Fees, and Cart Rates) and lists prices for Resident and Non-Resident players. A large blue arrow points from the website content towards the text on the right.

New Haven public golf course:

Segmenting the market by

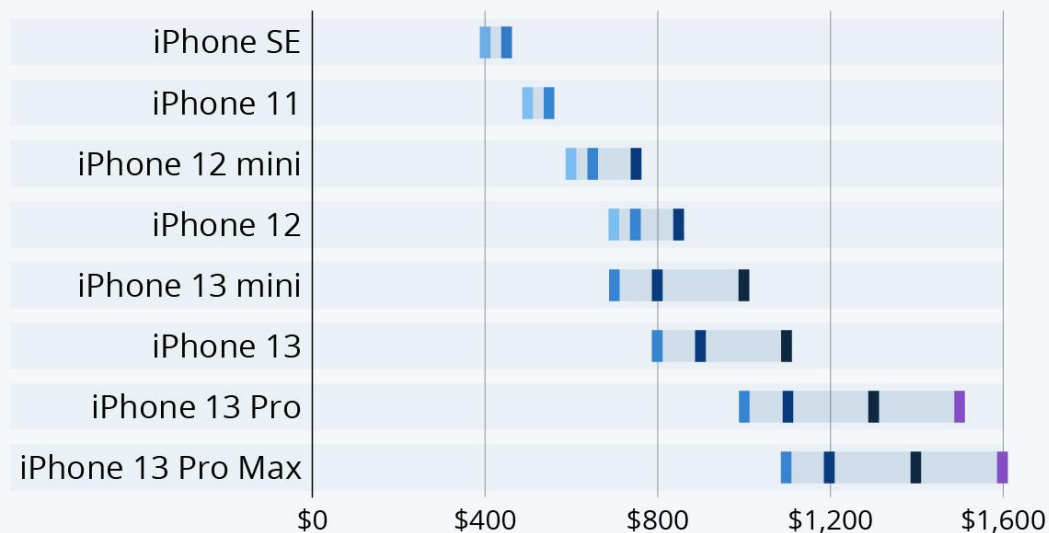
- time of day,
- day of week, and
- riding vs walking
- (and some verifiable demographic discrimination as well: local vs non, age)

Sure enough:

An iPhone for (Almost) Every Wallet

Pricing of Apple's current iPhone line-up in the United States

64GB 128GB 256GB 512GB 1TB



Source: Apple



statista

Alternative Self-Selecting Strategies: “Damaged Goods” Strategy

Example 1: Intel i486SX

- Intel introduces lower-end i486SX chip in 1991
- Chip is actually a high-end i486DX chip that has been intentionally damaged to disable the “floating point” unit
- Intel prices i486SX at half the price of the i486DX

Example 2: IBM LaserPrinter E

- IBM’s main printer was the LaserPrinter
- In 1990, IBM introduced the LaserPrinter E, which was identical except for an added chip that made the machine pause between printing pages
- Identical except printing speed

In both cases, the firm used a “damaged goods” strategy

- Priced “high quality” good well above cost to capture consumer surplus
- However, this left many consumers not buying the good, even though they value it above marginal cost.
- Solution: “damage” the good to create a low-quality version. Price it lower to capture additional consumers while still attracting high-value consumers to the high-quality product.

Think This is an Old Idea? Still Happening:



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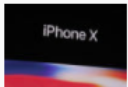
DISRUPT SF TechCrunch's flagship event begins today - It's not too late to get tickets [Get yours today ▶](#)

Tesla

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Tesla flips a switch to increase the range of some cars in Florida to help people evacuate

Posted Sep 9, 2017 by [Greg Kumparak \(@grg\)](#)



Tesla has pushed an over-the-air update to some of its vehicles in Florida that lets those cars go just a *liiiiittle* bit farther, thus helping their owners get that much farther away from the devastation of Hurricane Irma.

Wondering how that's even possible?

Up until a few months ago, Tesla sold a 60kWh version of its Model S and Model X vehicles — but the battery in those cars was *actually* rated at 75kWh. The thinking: Tesla could offer a more affordable 60kWh version to those who didn't need the full range of the 75kWh battery — but to keep things simple, they'd just use the same 75kWh battery and lock it on the software side. If 60kWh buyers found they needed more range and wanted to upgrade later, they could... or if Tesla wanted to suddenly bestow owners with some extra range in case of an emergency, they could.

And that's what's happening here.

As first noticed by [Tesla owners on Reddit](#), the company has pushed a "temporary update" to vehicles within the evacuation zones that bumps the 60kWh models up to 75kWh.

Crunchbase

Tesla

FOUNDED
2003

OVERVIEW

Tesla Motors was started by a group of Silicon Valley entrepreneurs and strives to create a revolution and accelerate the world's transition to electric mobility with a full range of increasingly affordable electric cars. Tesla vehicles are EVs (electric vehicles), which are transforming the way people drive and move. Tesla has gone public as of June 29, 2010 and has a market cap of \$34.32 billion. ...

LOCATION
Palo Alto, CA

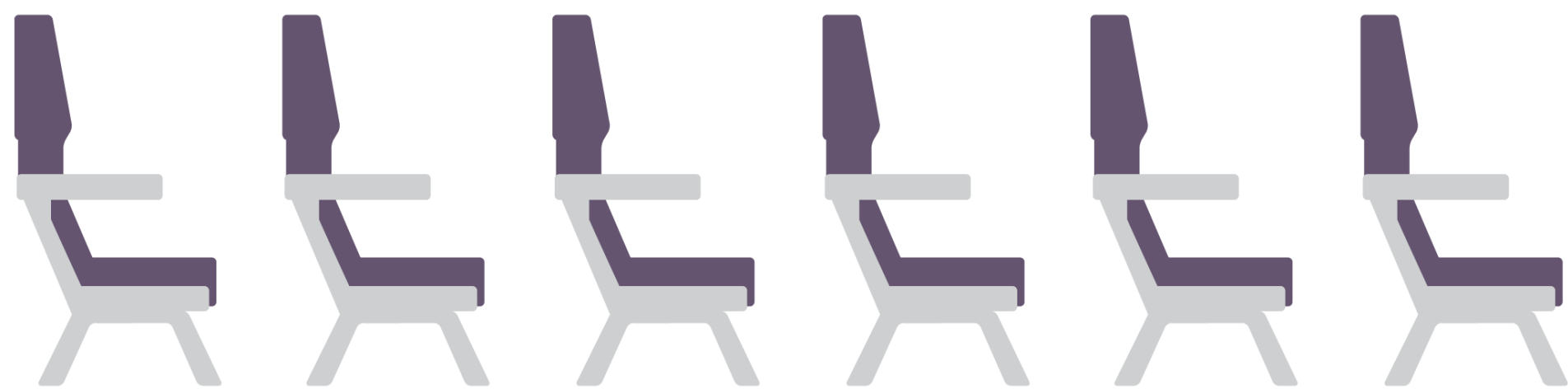
CATEGORIES
[Automotive](#), [Electronics](#), [Innovation Management](#)

Report: Airlines Installing Uncomfortable Bumps In Seatbacks Because It Pleases Them



NEW YORK—According to sources throughout the commercial aviation sector, the nation’s airlines will begin installing awkwardly placed bumps in every airplane seatback this week because it reportedly brings great pleasure to them. “Over the next four to six months, across our entire fleet, we’ll be rolling out seats with an irritating array of lumps and ridges to painfully jab our customers in the back, and we couldn’t be more delighted about it,” said United Airlines CEO Jeff Smisek, speaking on behalf of all international, regional, and low-cost carriers, every one of whom admitted to deriving immense joy from watching their passengers squirm and search in vain for a tolerable position. “Let me make clear that these seats will be incredibly uncomfortable, and there is nothing more gratifying than making the experience of simply sitting during one’s flight an excruciating ordeal.” Smisek noted that for an additional fee, customers would be allowed to purchase seats with only a single, pointed protrusion in the lower back. 

Do airlines do this?



Jet Blue

34"

Virgin America,
Southwest

32"

Delta,
American

31"

United

30"

Spirit

28"



Economy isn't the same everywhere. Why do many offer "Economy Plus"?

Airlines get this

FOR AIRLINES, A DELICATE BALANCE BETWEEN ECONOMICS AND COMFORT

Manufacturers are creating seats with an increasingly smaller profile, but instead of providing more legroom, airlines are using the space to squeeze in more rows. Aircraft design experts say the next evolution could be a standing "seat."

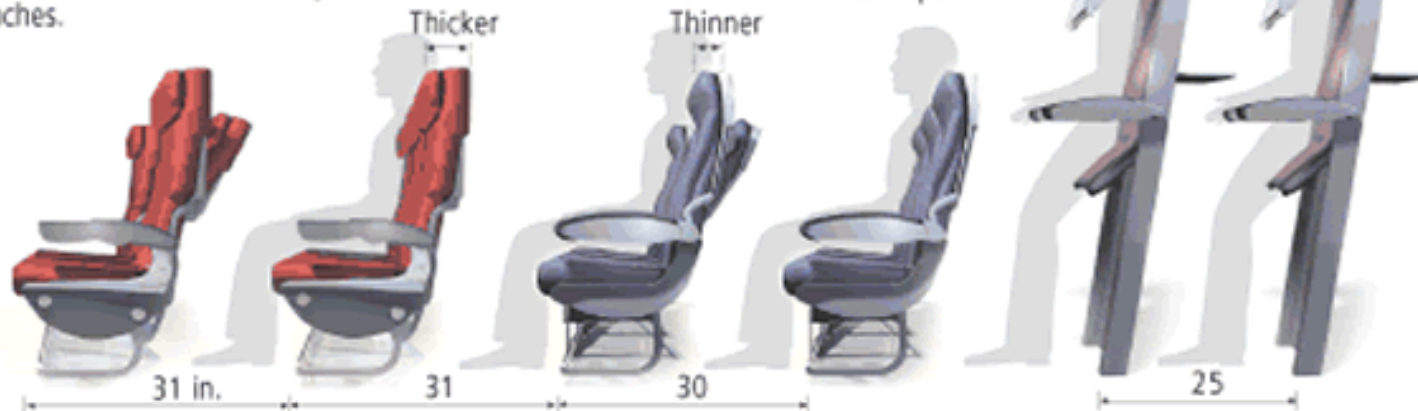
SEAT-TO-SEAT

Since 1978, the average space between economy-class seats has lost 3 inches.

Newer, thinner seats could have increased the legroom ...

...but many airlines chose to use the space to add more seats ...

...and Airbus has proposed a standing-room concept.



SIDE-BY-SIDE

Eight-abreast configuration

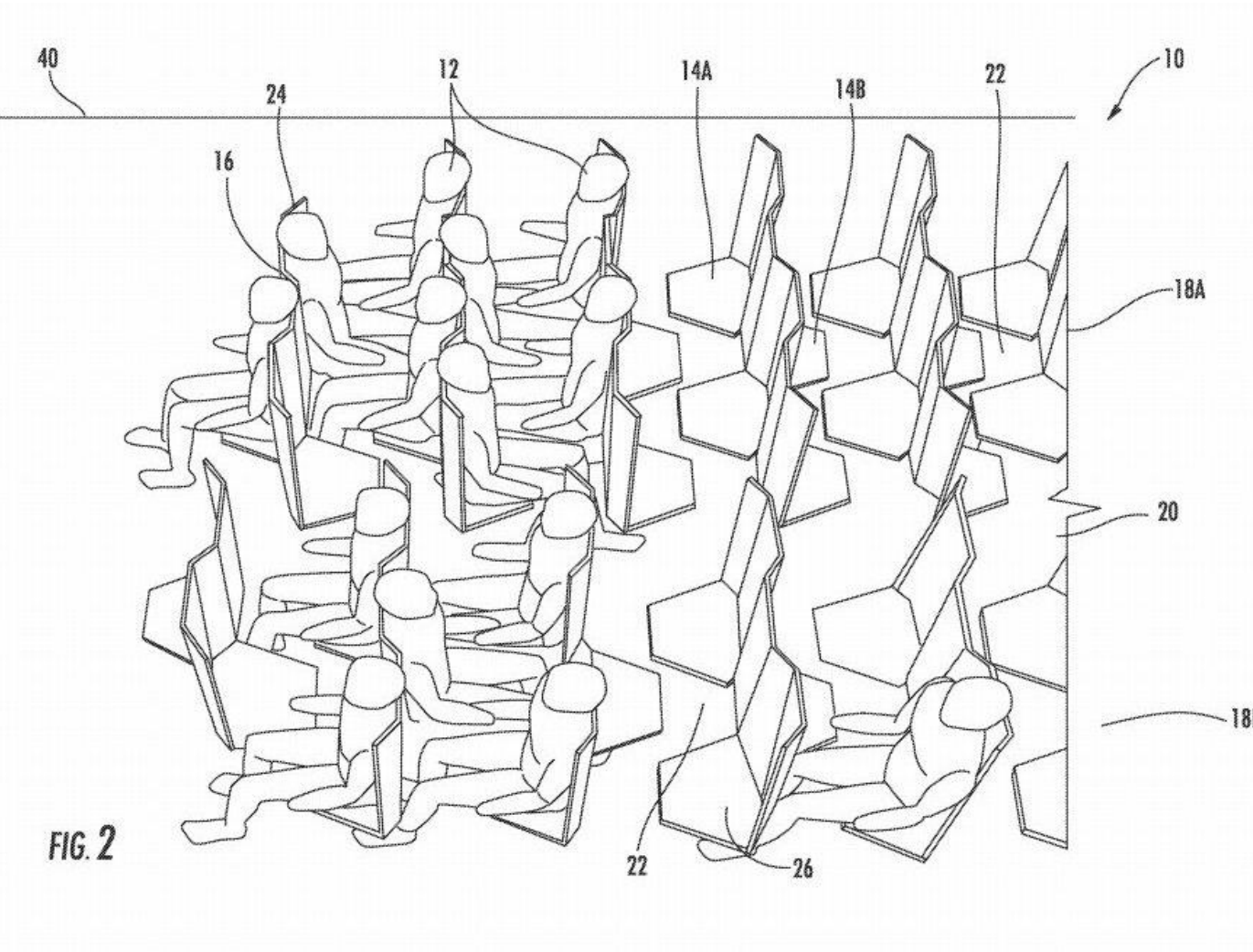
BOEING 787 "DREAMLINER"

Nine-abreast configuration

Boeing's recommended configuration for its 787 Dreamliner is eight seats across the aisle, which would give passengers some extra elbow room. More than half of the airlines ordering the plane, however, have opted for a nine-abreast configuration.

Think Thin!

More Innovation?



In theory...

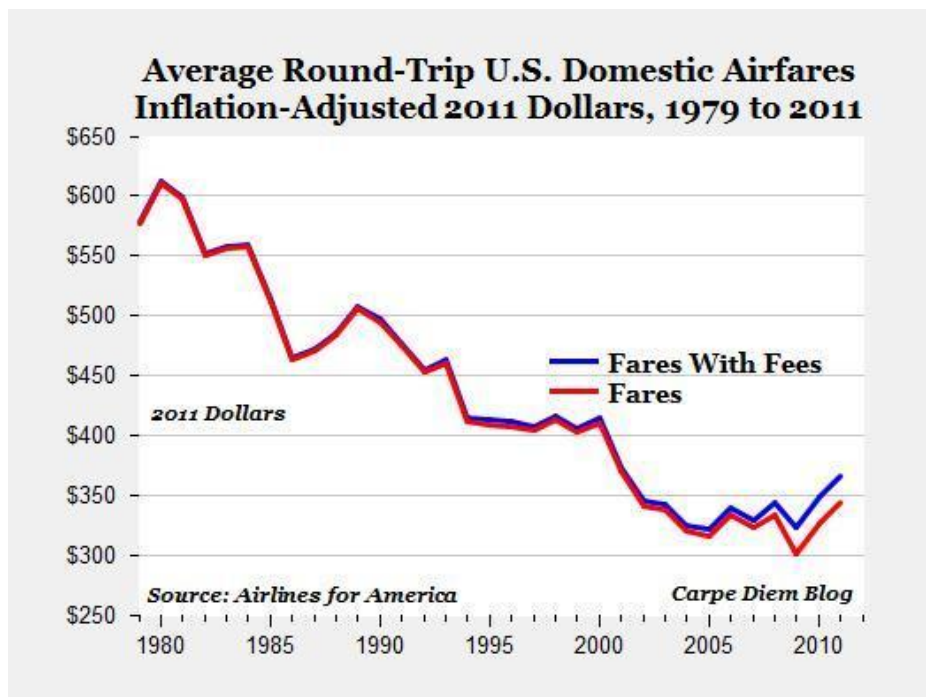
- Patent filed by Zodiac seating for “hexagon seating”
- Turns 6-across into 7-across on narrow-body aircraft.
- ... thoughts?

What does this say about our “constraints”?

Formerly, airline pricing was “one size fits all” – everything included in the airfare. Now, airlines recognize that people differ greatly in willingness-to-pay, and are engaging in price discrimination:

“Willing to Buy” constraint

- By breaking out fees, airlines can lower base airfares. This brings new people into the market who otherwise wouldn't have taken a flight.



- Spirit airlines is an ultra-low-cost airline.
- Who do they consider their competition? Greyhound.
- More people can afford to fly because of the fall in base fares.

What does this say about our “constraints”?

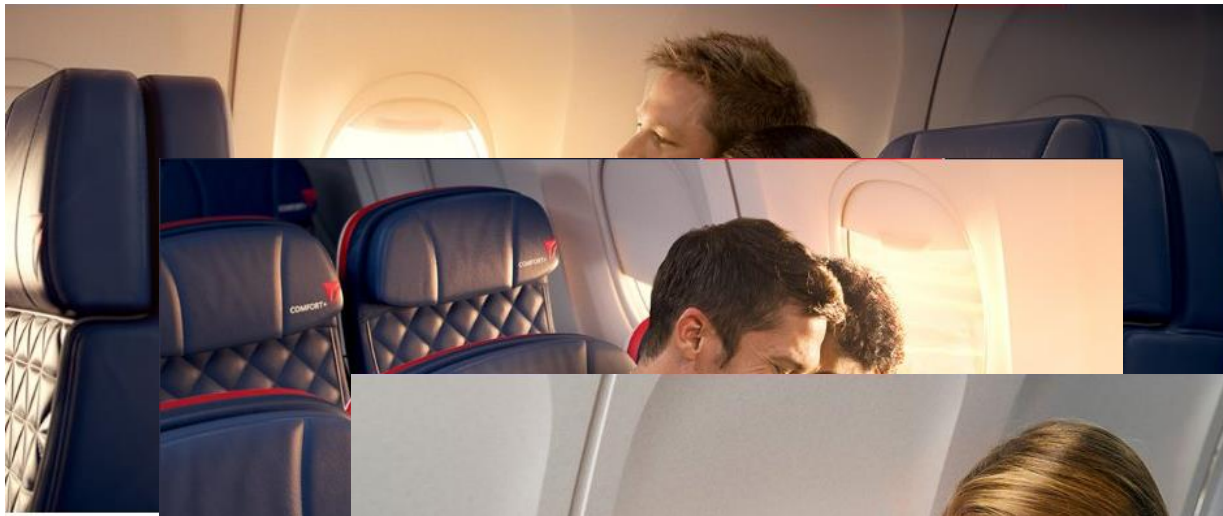
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“Willing to Buy” constraint

- By breaking out fees, airlines can lower base airfares. This brings new people into the market who otherwise wouldn’t have taken a flight.

“Buy the Right One” constraint

- Now there are add-ons targeted at different segments.
 - Board early.
 - Choose seats in advance.
 - Check bags.
 - Food.
- Can’t extract full surplus from these, but there is *a lot* of surplus around some of these in some segments.
- Think about the markups that result.



ON YOUR FLIGHT

An Elevated
Class

ON YOUR FLIGHT

Thoughtful
Comfort

A better experience
dedicated over

ON YOUR FLIGHT TO ATLANTA

A Higher Standard

On your Delta flight to Atlanta, enjoy amenities connected with available Wi-Fi and entertainment on seatback screen or your device. Customize your experience ahead of time.

ON YOUR FLIGHT TO ATLANTA

A Better Value Fare In Basic Economy

Enjoy the benefits of a Main Cabin experience at a value fare. Though some restrictions apply to your fare, you will still enjoy our award winning services in-flight, including

FROM
\$ 98.20

One Way Per Passenger

SELECT

Summary and direction

Today

Individual price discrimination

- Self-selecting: consider targeting high-end good to high types, low-end to low types. Need to make sure surplus leads high types to buy high-end.

Multi-unit Price Discrimination

- Individuals have downward-sloping demand curves over multiple units of some products
- Perfect quantity discrimination: charge exact valuations for subsequent units
- Membership fees: charge consumers their consumer surplus—be careful to cut them off at the quantity they value below marginal cost!

Next Lecture: Quantity AND individual price discrimination

- Menu pricing—self-selecting price discrimination where instead of quality of good, the differentiator is quantity
- Two-part tariffs—a price plans that involve a membership fee and a usage fee. This is a pricing plan that can be used for simple quantity price discrimination or individual *and* quantity price